

Quiz details

Quiz instructions:

This quiz will focus on the material from Week 6 in the lab workbook. To answer the following questions, you will need to download the dataset [old_forest_soil_pH.csv](https://canvas.stir.ac.uk/courses/19303/files/4997289/download?download_frd=1) ↓ (https://canvas.stir.ac.uk/courses/19303/files/4997289/download?download_frd=1), which includes estimates of soil pH in 16 locations of old growth forest in Scotland from 2010 and 2020. Note that there is one missing value from 2020 in location 8.

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Old forest soil data

To answer the following questions, you will need to download the dataset [old_forest_soil_pH.csv](https://canvas.stir.ac.uk/courses/19303/files/4997289?wrap=1) (<https://canvas.stir.ac.uk/courses/19303/files/4997289?wrap=1>), which includes estimates of soil pH in 16 locations of old growth forest in Scotland from 2010 and 2020. Note that there is one missing value from 2020 in location 8.

Question 1 1 pts

Suppose that you want to test the null hypothesis that the mean pH from 2010 is less than 4.8. Based on a Shapiro-Wilk test of normality and Q-Q plot, what would be the most appropriate statistical test to use?

- ☐ One sample t-test
- ☐ Independent samples t-test
- ☐ Paired samples t-test
- ☐ Wilcoxon test
- ☐ Mann-Whitney U test

Correct answer comments

Because we want to test if the mean of a sample is different from a specific value (4.8), we need to use either a one sample t-test or a Wilcoxon test. We can eliminate the independent samples t-test, the paired samples t-test, and the Mann-Whitney U test as options because these are all used to test for differences between two different groups (e.g., if we wanted to test for differences between years). This leaves the one sample t-test and the Wilcoxon test as options. We should use a one sample t-test if our data are normally distributed and a Wilcoxon test if the data are not normally distributed. To test whether or not the data are normally distributed, we need to use the **Assumption Checks** option in Jamovi. From the Jamovi toolbar, navigate to 'T-Tests' then select 'One Sample T-Test' from the pulldown menu. Put 'pH_2010' into the Dependent variables box and then check the box for a Normality test and Q-Q plot. Since the data appear to be roughly along the diagonal line of the Q-Q plot, and we do not reject the null hypothesis of our normality test ($P = 0.690$), we can conclude that the data are normally distributed. We can therefore use the one sample ("Student's") t-test.

Wrong answer comments

Because we want to test if the mean of a sample is different from a specific value (4.8), we need to use either a one sample t-test or a Wilcoxon test. We can eliminate the independent samples t-test, the paired samples t-test, and the Mann-Whitney U test as options because these are all used to test for differences between two different groups (e.g., if we wanted to test for differences between years). This leaves the one sample t-test and the Wilcoxon test as options. We should use a one sample t-test if our data are normally distributed and a Wilcoxon test if the data are not normally distributed. To test whether or not the data are normally distributed, we need to use the **Assumption Checks** option in Jamovi. From the Jamovi toolbar, navigate to 'T-Tests' then select 'One Sample T-Test' from the pulldown menu. Put 'pH_2010' into the Dependent variables box and then check the box for a Normality test and Q-Q plot. Since the data appear to be roughly along the diagonal line of the Q-Q plot, and we do not reject the null hypothesis of our normality test ($P = 0.690$), we can conclude that the data are normally distributed. We can therefore use the one sample ("Student's") t-test.

Question 2 1 pts

Now run the appropriate test from Question 1. What is the test statistic? Report your answer to 2 decimal places.

-3.28

Correct answer comments

From the Jamovi toolbar, navigate to 'T-Tests' then select 'One Sample T-Test' from the pulldown menu. Put 'pH_2010' into the Dependent variables box and then check the box for "Student's" under the **Tests** options. Under **Hypothesis**, set the Test value to 4.8, then choose "< Test value" (the third radio button down). This will test the null hypothesis that the mean of pH_2010 is less than 4.8. From the table to the right titled "One Sample T-Test", you can find the test statistic '-3.284', which can be rounded to 2 decimals as '-3.28'.

Wrong answer comments

From the Jamovi toolbar, navigate to 'T-Tests' then select 'One Sample T-Test' from the pulldown menu. Put 'pH_2010' into the Dependent variables box and then check the box for "Student's" under the **Tests** options. Under **Hypothesis**, set the Test value to 4.8, then choose "< Test value" (the third radio button down). This will test the null hypothesis that the mean of pH_2010 is less than 4.8. From the table to the right titled "One Sample T-Test", you can find the test statistic '-3.284', which can be rounded to 2 decimals as '-3.28'.

Question 3 1 pts

From the test run in Question 1 and 2, what can you conclude?

- ☐ Fail to reject the null hypothesis that the mean pH in 2010 was 4.8 because $P > 0.05$
- ☐ Fail to reject the null hypothesis that the mean pH in 2010 was 4.8 because $P < 0.05$
- ☐ Reject the null hypothesis that the mean pH in 2010 was 4.8 because $P > 0.05$
- ☐ Reject the null hypothesis that the mean pH in 2010 was 4.8 because $P < 0.05$
- ☐ The test results are inconclusive

Question 4 1 pts

Suppose that you want to test the null hypothesis that the mean pH from 2020 is 4.8

against the alternative hypothesis that mean pH from 2020 is less than 4.8. Find the appropriate statistical test to use, then run it in Jamovi. What is the correct test statistic, p-value, and conclusion?

- ☐ $t = 1.357$, $P = 0.902$; do not reject null hypothesis.
- ☐ $t = 1.357$, $P = 0.902$; reject null hypothesis.
- ☐ $W = 73.00$, $p = 0.909$; do not reject null hypothesis.
- ☐ $W = 73.00$, $p = 0.909$; reject null hypothesis.
- ☐ None of the above

Correct answer comments

From the Jamovi toolbar, navigate to 'T-Tests' then select 'One Sample T-Test' from the pulldown menu. Put 'pH_2020' into the Dependent variables box. Run the **Assumption Checks** of normality, then note that the null hypothesis that the data are normally distributed should be rejected ($P = 0.018$). It is therefore necessary to use a Wilcoxon rank test instead of a Student's t-test. Check the box for "Wilcoxon rank" under the **Tests** options. Under **Hypothesis**, set the Test value to 4.8, then choose "< Test value" (the third radio button down). This will test the null hypothesis that the mean of pH_2020 is less than 4.8. From the table to the right titled "Wilcoxon W", you can find the test statistic 73.00 and the p-value 0.909. Because $P > 0.05$, we do not reject the null hypothesis.

Wrong answer comments

From the Jamovi toolbar, navigate to 'T-Tests' then select 'One Sample T-Test' from the pulldown menu. Put 'pH_2020' into the Dependent variables box. Run the **Assumption Checks** of normality, then note that the null hypothesis that the data are normally distributed should be rejected ($P = 0.018$). It is therefore necessary to use a Wilcoxon rank test instead of a Student's t-test. Check the box for "Wilcoxon rank" under the **Tests** options. Under **Hypothesis**, set the Test value to 4.8, then choose "< Test value" (the third radio button down). This will test the null hypothesis that the mean of pH_2020 is less than 4.8. From the table to the right titled "Wilcoxon W", you can find the test statistic 73.00 and the p-value 0.909. Because $P > 0.05$, we do not reject the null hypothesis.

Question 5 1 pts

If we somehow knew that the **true mean value** of pH in 2020 was actually less than 4.8, then what could we conclude about the statistical analysis from Question 4?

- ☐ The analysis was correct
- ☐ There was a Type I error
- ☐ There was a Type II error
- ☐ There was a Type III error
- ☐ There was either a Type I or Type II error, but we cannot know which one

Correct answer comments

In Question 4, we should have failed to reject the null hypothesis. A Type I error occurs when we reject the null hypothesis but it is actually true. A Type II error occurs when we fail to reject the null hypothesis but it is actually false. A Type III hypothesis is not something that we have learned (it is sometimes defined as a situation in which the null hypothesis is correctly rejected, but for the

wrong reason). If the true mean pH from 2020 was less than 4.8, and we failed to reject the null hypothesis, then we would have a Type II error.

General answer comments

In Question 4, we should have failed to reject the null hypothesis. A Type I error occurs when we reject the null hypothesis but it is actually true. A Type II error occurs when we fail to reject the null hypothesis but it is actually false. A Type III hypothesis is not something that we have learned (it is sometimes defined as a situation in which the null hypothesis is correctly rejected, but for the wrong reason). If the true mean pH from 2020 was less than 4.8, and we failed to reject the null hypothesis, then we would have a Type II error.

Question 6 1 pts

Suppose that you now want to test the hypothesis that the soil has become more acidic (i.e., that the pH has become lower) from 2010 to 2020. What would be the most appropriate statistical test to use to test the null hypothesis in this case?

- ☐ One sample t-test
- ☐ Independent samples t-test
- ☐ Paired samples t-test
- ☐ Wilcoxon test
- ☐ Mann-Whitney U test

Correct answer comments

The data for this test are naturally paired by location. That is, we want to know if the change in pH between 2010 and 2020 is negative, on average, for different locations. From the Jamovi toolbar, navigate to 'T-Tests' then select 'Paired Samples T-Test' from the pulldown menu. Put 'pH_2010', then 'pH_2020' into the Dependent variables box. Do not forget to run the **Assumption Checks** to see if the differences between years are normally distributed! From the test of normality (Shapiro-Wilk test), we can see that there is no reason to reject the null hypothesis that the data are normally distributed ($P = 0.475$). We can therefore use a Student's t-test instead of the nonparametric Wilcoxon test.

Wrong answer comments

The data for this test are naturally paired by location. That is, we want to know if the change in pH between 2010 and 2020 is negative, on average, for different locations. From the Jamovi toolbar, navigate to 'T-Tests' then select 'Paired Samples T-Test' from the pulldown menu. Put 'pH_2010', then 'pH_2020' into the Dependent variables box. Do not forget to run the **Assumption Checks** to see if the differences between years are normally distributed! From the test of normality (Shapiro-Wilk test), we can see that there is no reason to reject the null hypothesis that the data are normally distributed ($P = 0.475$). We can therefore use a Student's t-test instead of the nonparametric Wilcoxon test.

Question 7 1 pts

Run the most appropriate statistical test identified in Question 6. If the null hypothesis is true and pH has not decreased from 2010 to 2020, then what is the probability of getting a test statistic as low as the one calculated from the test?

Report your answer to 3 decimal places.

0.003

Correct answer comments

From the Jamovi toolbar, navigate to 'T-Tests' then select 'Paired Sample T-Test' from the pulldown menu. Put 'pH_2010', then 'pH_2020' into the Dependent variables box and then check the box for "Student's" under the **Tests** options. Under **Hypothesis**, choose "Measure 1 < Measure 2" (the third radio button down). From the table to the right titled "Paired Samples T-Test", you can find the test statistic '-3.234' and the corresponding p-value 0.003. Remember that the p-value is the probability of getting a result as or more extreme than the one observed assuming that the null hypothesis is true, so this is what you should report.

Wrong answer comments

From the Jamovi toolbar, navigate to 'T-Tests' then select 'Paired Sample T-Test' from the pulldown menu. Put 'pH_2010', then 'pH_2020' into the Dependent variables box and then check the box for "Student's" under the **Tests** options. Under **Hypothesis**, choose "Measure 1 < Measure 2" (the third radio button down). From the table to the right titled "Paired Samples T-Test", you can find the test statistic '-3.234' and the corresponding p-value 0.003. Remember that the p-value is the probability of getting a result as or more extreme than the one observed assuming that the null hypothesis is true, so this is what you should report.

Question 8 1 pts

Now, suppose that you want to test the null hypothesis that the average 2020 soil pH is the same in North and South areas. Based on this dataset, which of the following tests should you run?

- ☐ Student's independent samples t-test
- ☐ Welch's independent samples t-test
- ☐ Mann-Whitney U test
- ☐ Wilcoxon test
- ☐ Paired samples t-test

Correct answer comments

Here we need to test whether or not there is a difference between 2 independent groups, so we can eliminate the Wilcoxon test and the Paired samples t-test, which do not test for differences between independent groups. The choice between the Student's t-test, Welch's t-test, and the Mann-Whitney U test comes down to the **Assumption Checks**. If we reject the null hypothesis that the data are normally distributed, then we should use the Mann-Whitney U test. There is no reason to reject this null hypothesis ($P = 0.179$), so we do not use the Mann-Whitney U test. Next, if we reject the null hypothesis that the groups have equal variances (Levene's test), then we need to use the Welch's t-test. There is no reason to reject the null hypothesis that the groups have equal variances ($P = 0.085$), so we can continue with the Student's t-test.

Wrong answer comments

Here we need to test whether or not there is a difference between 2 independent groups, so we can eliminate the Wilcoxon test and the Paired samples t-test, which do not test for differences

between independent groups. The choice between the Student's t-test, Welch's t-test, and the Mann-Whitney U test comes down to the **Assumption Checks**. If we reject the null hypothesis that the data are normally distributed, then we should use the Mann-Whitney U test. There is no reason to reject this null hypothesis ($P = 0.179$), so we do not use the Mann-Whitney U test. Next, if we reject the null hypothesis that the groups have equal variances (Levene's test), then we need to use the Welch's t-test. There is no reason to reject the null hypothesis that the groups have equal variances ($P = 0.085$), so we can continue with the Student's t-test.

Question 9 1 pts

How many degrees of freedom are there in the test identified in Question 8? Report your answer as an integer value.

13

Correct answer comments

We can find this value under 'df' in the Independent Samples T-test table. Note that the value given is 13.00, which is '13' when reported as an integer.

Wrong answer comments

We can find this value under 'df' in the Independent Samples T-test table. Note that the value given is 13.00, which is '13' when reported as an integer.

Question 10 1 pts

To what null distribution is the test statistic be compared in the test run in Question 8?

- ☐ Normal distribution
- ☐ Binomial distribution
- ☐ Poisson distribution
- ☐ Uniform distribution
- ☐ t-distribution
- ☐ F distribution

Correct answer comments

The t-test compares the test statistic 2.096 to the **t-distribution**, then finds the probability of getting this test statistic to be $P = 0.056$ if the t statistic was in fact calculated from data having the same mean between groups.

Wrong answer comments

The t-test compares the test statistic 2.096 to the **t-distribution**, then finds the probability of getting this test statistic to be $P = 0.056$ if the t statistic was in fact calculated from data having the same mean between groups.